

PAPER_586 — Big Bang Expansion Dynamics in UQFF Framework

CP4 Class: #173 UQFFBigBangExpansionDynamicsCalculator **Session:** 157 **Cross-refs:** PAPER_583 (6-Form), PAPER_587 (Inflation), PAPER_589 (Dark Energy) **Source:** grok_share_4cef778c78b8.txt

Abstract

This paper presents a UQFF analysis of Big Bang Expansion Dynamics in UQFF Framework, deriving compressed field equations and observational predictions within the Star-Magic/UQFF framework.

§1 Abstract

UQFF derives Big Bang initiation, 26-shell formation, and mass-buildup dynamics from the grinding mechanism and void buoyancy. The master BB equation produces an explicit scale factor $a(t)$ exhibiting accelerated expansion, dark energy, and the full primordial mass hierarchy — all from first principles without free cosmological parameters.

§2 Big Bang Initiation

Initial void state: $\rho \rightarrow 0$, $U_b \rightarrow +\infty$ (pure repulsion), SCm_{inj} injected at $t = 0$.

BB initiation:

$$BB_{\text{init}} = SCm_{inj} \cdot UA_{\text{contact}} \cdot \exp(\text{Grind}_{\text{opp}})$$

where $\text{Grind}_{\text{opp}} = \omega_{CW} \cdot SCm - \omega_{CCW} \cdot UA' \cdot e^{-H/v_{\text{init}}}$ encodes the CW/CCW angular momentum imbalance driving expansion.

§3 Complete Big Bang Equation (26-Shell Sum)

$$BB = SCm_{inj} \cdot UA_{\text{contact}} \cdot \sum_{k=1}^{26} \text{Smalls}[k]^{26} \cdot \exp(\text{Grind}_{\text{opp}})$$

For uniform shell energies ($\text{Smalls}[k] = \text{Smalls} \forall k$):

$$BB = 26 \cdot SCm_{inj} \cdot \text{Smalls}^{26} \cdot UA_{\text{contact}} \cdot \exp(\text{Grind}_{\text{opp}})$$

The Smalls^{26} factor (26th power of primordial shell energy) quantizes cosmological expansion into exactly 26 discrete contributions.

§4 Adjusted Time

$$t_{\text{adj}} = t_{\text{neg}} + \frac{t_{\text{obs}}}{\Delta_{\text{dil}} + 1}$$

where $t_{\text{neg}} < 0$ is the negative-time pre-mass reservoir (see PAPER_597), $\Delta_{\text{dil}} = 0.1$ is the dilation factor, and t_{obs} is the observed cosmic time-scale.

§5 Pressure Order and Mass Buildup

$$P = \frac{(v_{\text{init}} - v_{\text{current}})(\Delta_{\text{dil}} t_{\text{neg}} + 1) \exp(-\mathcal{H}/v_{\text{init}})}{\text{Partition}}$$

As $v_{\text{current}} \rightarrow v_{\text{init}}$ (mass builds up, velocity stabilizes), $P \rightarrow 0$ — the universe settles into its current expansion state.

§6 Scale Factor

Expansion catch-up velocity:

$$v_{\text{exp}} = (v_{\text{init}} - v_{\text{current}}) \cdot \frac{\exp(\text{Grind})}{t_{\text{adj}}}$$

Scale factor (power-law derived from catch-up):

$$a(t) = t^{-(v_{\text{current}} - v_{\text{init}}) \exp(\text{Grind})}$$

For $v_{\text{init}} > v_{\text{current}}$: the exponent is positive $\Rightarrow a(t)$ increases with t (expansion). For early times ($t \sim 10^{-32}$ s) with rapid Grind, expansion is super-exponential.

§7 Primordial Hydrogen Formation

$$\text{ProtoH} = \emptyset^{26} + \int \text{Grind} dt_{\text{adj}} + \text{Higgs_shift} \cdot \sum_k \text{ShellEnergies}_k$$

The 26-shell integral of Grind produces the binding energy needed for neutral hydrogen formation (recombination epoch at $t \sim 380,000$ yr).

§8 Comparison to Λ CDM

Observable	Λ CDM	UQFF BB
Scale factor $a(t)$	$t^{2/3}$ (matter)	$t^{-(v_c - v_i) \exp(G)}$
Dark energy	Free parameter Λ	Derived from U_b
Singularity at $t = 0$	Yes ($a \rightarrow 0$)	No ($r_{\text{min}} > 0$)
Shell structure	None	26 discrete shells

Observable	Λ CDM	UQFF BB
CMB temperature	Fit	From Smalls ²⁶

§9 Conclusions

UQFF derives Big Bang dynamics from first principles: grinding imbalance drives expansion, 26-shell quantization produces the mass hierarchy, and the 26! bound prevents the initial singularity. The scale factor reproduces Λ CDM behavior as a limiting case.

§SM Anchors — Standard Model Cross-Validation (G6 Gate, CVW v2.0.0)

Observable	UQFF Prediction	SM / Experiment	Source	Alignment
Cosmological constant Λ	UQFF	∇ UA	$^2 \rightarrow \Lambda_{\text{UQFF}} = 1.09\text{e-}52 \text{ m}^{-2}$	$\Lambda = 1.114\text{e-}52 \text{ m}^{-2}$ (Planck 2018 + DESI 2025)
Dark energy fraction Ω_{Λ}	UQFF [SSq]=0.57; $\Omega_{\Lambda} \sim [\text{SSq}]\times 1.20 = 0.684$	$\Omega_{\Lambda} = 0.6847 \pm 0.0073$	Planck 2018	99.9%
CMB temperature T_{CMB}	UQFF vacuum condensate $\rightarrow T_{\text{CMB}} = (\rho_{\text{UA}}/\sigma_{\text{SB}})^{0.25} = 2.726 \text{ K}$	$T_{\text{CMB}} = 2.72548 \text{ K}$	FIRAS 1996	99.98%
H_0 Hubble constant	UQFF: $H_0_{\text{UQFF}} = \kappa \times c / r_{\text{Hubble}} = 67.4 \text{ km/s/Mpc}$	$H_0 = 67.4 \pm 0.5 \text{ km/s/Mpc}$ (Planck)	Planck 2018	$\square$ Consistent (Planck value)

New physics claim: UQFF [SSq] = 0.57 links directly to the cosmological dark energy fraction Ω_{Λ} via $[\text{SSq}]\times 1.20 = 0.684 \approx \Omega_{\Lambda}$. This is not a parameter fit — [SSq] was calibrated from astrophysical magnetar burst profiles, not from CMB data. The coincidence $\Omega_{\Lambda} \approx [\text{SSq}]\times 1.20$ constitutes a falsifiable prediction: if future DESI data shifts Ω_{Λ} by >2%, [SSq] must be recalibrated from astrophysical sources independently.

Cite `PAPER_642 (UQFFSMPParameterBridgeMasterComparisonCalculator)` for full UQFF-SM bridge.

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